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Task Allocation in Human-Robot Collaboration (HRC) Based on Task Characteristics and Agent Capability for Mold Assembly

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Abstract

The involvement of repetitive motion with force and heavy component handling during the manual mold assembly causes an ergonomic problem that harms the musculoskeletal system on the human worker. This ergonomic problem can be eliminated using a robot. However, the traditional automation system is impractical to be implemented in the mold assembly because of the complexity of the customized mold structure. Hence, human-robot collaboration (HRC) systems became the potential solution to improve the working conditions with the assistance of a robot while adapting to the frequent change over in the mold assembly operation. In this paper, we propose a task allocation model for the HRC system that consists of three agents who are one human and two robots to cope with heavy part handling and the diversity of tasks in the mold assembly. First, we decompose the assembly operation into functional actions. Second, we evaluate the action assignment using the Analytic Network Process (ANP) based on the action characteristics and agent capability. Third, we allocate the functional action to the agents in the HRC work cell using the Genetic Algorithm (GA) to minimize the operation cycle time and maximize the agent capability. The outcome of this paper is to contribute to the expansion of the HRC system with one human and two robots in the mold assembly operation. In addition to the improvement in the ergonomic condition, the proposed task allocation model can achieve a wider variety of tasks assigned to robots with minimum tool change.

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Keywords: Human-robot collaboration; Task Allocation; Work decomposition; Analytic Network Process (ANP); Genetic Algorithm

1. Introduction

Traditional automation systems that make use of industrial robots have been widely used in manufacturing operations especially for mass production since the past few decades. However, the manufacturing paradigm has been shifted from mass production to mass customization and further to mass personalization which requires high flexibility and adaptability of automation systems to deal with the rapid changes in customer demand, product and process [1]. As a result of the evolution, traditional automation systems are facing challenges to adapt to the new manufacturing environment. These challenges include the limitation in automating every task performed in a manufacturing environment and quick changing in the configuration of work cell due to the cost and

technological limitation. In order to overcome the limitations in traditional automation systems, human-robot collaboration (HRC) systems became the potential solution in order to achieve a maximum level of automation while providing flexibility to cope with the changes in production.

An HRC system is a system where a human and robot work together on the same task within a shared workspace. Humans have excellent cognitive ability and adaptability. Robots are good at performing repetitive tasks more efficiently with a consistent level of accuracy. The combination of human and robot's ability can improve flexibility [2] and ergonomics factors [3] in manufacturing operations. Furthermore, the product quality of the HRC system can be improved by integrating a computer vision system for inspection and part recognition purpose [4]. HRC systems for assembly have been

implemented in various industries such as automotive and aircraft [5,6], electronics [7,8] and mechanical assembly [9,10]. However, the application of HRC systems in the tool and die industry such as mold industry is still limited. These industries seek for the implementation of a certain level of automation because of the decrease in the number of skilled workers needed to perform the manual mold assembly. The mold assembly is the critical procedure in the mold making that requires comprehensive understanding of mold structure to ensure the functionality of the mold. Besides, the manual mold assembly involves heavy material handling and repetitive tasks that required force exertion. These tasks can cause musculoskeletal disorder (MSD) risk on the human worker especially on the upper limbs. Although the use of an industrial robot can relieve the MSD risk, but it will limit the intercept of the human during the operation because of the safety fence. The collaboration between the human and the robot is required to perform tight tolerance assembly and to understand the customized mold design. Thus, the use collaborative robot (cobot) that equipped with sensor is preferred to ensure the safety of the human worker during the collaborative operation.

Most of the HRC systems in the existing applications consist of one human and one cobot. However, this kind of HRC system is inadequate to meet the needs of a mold assembly. The HRC system for mold assembly must be able to handle the variety and weight of components in a mold. Therefore, we propose an HRC system that consists of one human and two cobots in this research. The objectives of this paper are to propose three collaboration modes and to develop a task allocation model for collaboration of one human with two robots (1H:2R) in the mold assembly operation. Our model aims to maximize the capabilities of agents and minimize the cycle time of the mold assembly operation. The expected contribution of this paper is the task allocation model that allocates tasks to a human and robots based on their capabilities and the characteristics of tasks.

Background and related research of this study is presented in Section 2. Section 3 describes the framework of the proposed task allocation model; while Section 4 explains the implementation of genetic algorithm. Then, a case study for the implementation of the model proposed and discussion are presented in Section 5, and finally Section 6 concludes this paper and lists the future work.

2. Literature Review

2.1. Task allocation in HRC system

Task allocation is one of the most important steps in the implementation of the HRC system. An efficient task allocation ensures the safety of the human and the optimum performance of a human and a robot during the collaborative operation. Mateus et al. [11] decomposed assembly operation to six levels hierarchy by using the information from CAD model. The top level is operation, then followed by subassembly, task, function, function stage, and motion levels. Resource capability assessment is then performed to identify operator support mode. Nikolakis et al. [12] broke down the collaborative jobs between a human and robot into two levels

hierarchy model that composed of task and operation. They then implemented a multi-criteria decision-making framework for offline task allocation and online re-scheduling. Another framework for task allocation alternative plans generation using resource suitability, resource availability and operation was proposed by Tsarouchi et al. [13,14]. Bruno and Antonelli [9] developed a method for task assignment in HRC work cell using task classification to classify the tasks to four classes, which are executable only by a human, executable only by a robot, executable only by human or robot, executable mandatorily by both a human and a robot. The classified task is then assigned based on the availability of the human or robot. Heydaryan et al. [15] applied Analytic Hierarchy Process (AHP) to identify the potential of human and robot collaboration for each task and Hierarchical Task Analysis (HTA) to allocate operational tasks to human and robot. As a conclusion from the existing researches, the criteria used to analyze resource suitability can be characterized into workspace factor, robot's reachability [13,14,16], human factors [11,15,17] and, part characteristics such as weight and dimension [9,18]. In addition, most of the researchers evaluated performances of the human-robot collaboration system based on waiting time, production time, mean flow time and cost [18–20].

The researches mentioned above are mainly focused on task allocation based on criteria instead of the optimization problem. For the optimization problem in an HRC system, Bänziger et al. [24] optimized waiting time and traveling distance in an HRC workplace. Dalle Mura and Dini [23] focused on minimization of cost, the number of skilled workers, and energy load variance in an assembly line. Both pieces of research implemented a Genetic Algorithm in their optimization problems. In summary, the existing researches related to task allocation in an HRC system focused only on either resource analysis based task allocation or optimization of task allocation. Thus, in this paper, we perform the task and resource analysis, then input the result from the analysis into the optimization problem. We divide the analysis criteria into skill and part requirements. The skill requirement indicates the skill needed to perform the task and the part requirement indicates the part characteristics in each task. We then perform the task allocation by applying the Genetic Algorithm to optimize the production time based on the outcome of the ANP analysis and tool change requirements.

2.2. Multi-robot HRC systems

Generally, the configurations of the HRC system considered in the existing researches are composed of one human and one robot. Most of the methodology in the existing research focused on deciding only one suitable resource for task allocation based on the robot's capability. If the resource is unavailable to perform the assigned task, then there will be a delay in the production. Besides, those tasks that cannot be performed by the robot will assign to the human. Hence, it is inefficient to reduce the ergonomic risk on the human. Sadrfaridpour, Saeidi, and Wang [21] mentioned briefly about multiple human-robot interaction levels in an HRC system. The workspaces of each robot are separated at the medium

interaction level. At the high interaction level, the robots can interact with each other and humans. Besides, Ding, Schipper, and Matthias [22] studied the collaboration between two humans with a dual arm-robot. Dalle Mura and Dini [23] presented an assembly line consists of twelve HRC workstations for the application in heavy industry. In this paper, we propose the collaboration of one human with two robots (1H:2R) to meet the characteristics of the mold assembly operation.

3. Framework of Task Allocation in HRC System

Fig. 1 shows the framework of the proposed task allocation model for human-robot collaboration in the mold assembly based on task characteristics and capability of agents in the HRC work cell. In this paper, we define an agent as the resource to execute the assigned functional action. An agent can be a human, a robot, both robots, or a human with a robot. The three levels in the proposed task allocation model are task decomposition, functional action-agent assignment priority analysis, and allocation of functional actions. Details of each level are discussed in the following sub-sections.

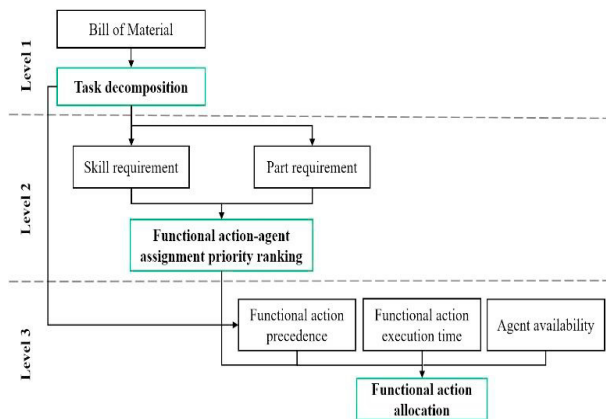


Fig. 1. Framework of task allocation model.

3.1. Task decomposition

Fig. 2 shows the hierarchy of task decomposition in the task allocation model. We decompose the assembly operation into sub-assemblies and further decompose each sub-assembly into a series of tasks. The simplest element in the decomposition model is the functional action of each task. In this paper, we define a functional action as an action that is performed on a specific part.

For the ease of analysis, we categorize functional actions in the mold assembly into nine types as shown in Table 1. A short code with two characters is used to represent each functional action type. The first character describes the action performed. "H" represents lift and position plate; "L" represents pick and locate component; "T" represents tightening action, and "F" represents action with force exertion. The second character represents tolerance level or type of component. "R", "F", and "T" describe the rough, fair, and tight tolerance level respectively. "S" and "P" at the second character represent

small component such as screw and plate respectively. Hence, for example, HF means to lift and position plate with fair tolerance, and FS means to insert small components with force exertion.

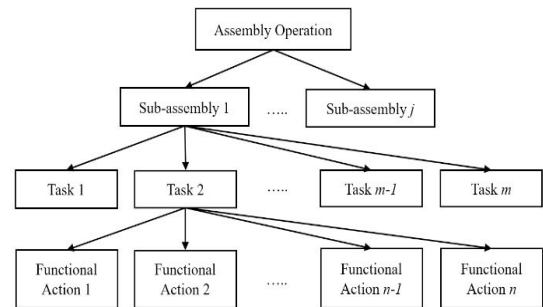


Fig. 2. Task decomposition.

3.2. Functional action-agent assignment priority using ANP

In the proposed framework, we use the Analytic Network Process (ANP) to decide the priority of the agent assignment for each functional action. The agents available in the HRC work cell are ranked based on the task characteristics in the mold assembly and resource's capability. Although Analytic Hierarchy Process (AHP) is common for multi-criteria decision making, but AHP is not suitable for our HRC assignment problem. AHP constructs the problem into hierarchy that consists of four levels which are goal, criteria, sub-criteria, and alternatives. However, AHP only considers the influence from top level to the next level and does not consider the influences between criteria. In contrast, ANP considers criteria and alternatives as nodes in a network. Thus, we can use ANP to model the influences and feedback between the nodes. Since the decision criteria included for the HRC assignment priority has influences on each other and on the alternatives, hence ANP is applied instead of AHP [25].

The designed ANP network (Fig. 3) composes of four clusters. The assignment cluster contains a type of functional action. Skill and part requirements are the criteria that indicate the characteristics of the functional action as well as the capability of agents. The nodes in the skill cluster are ergonomics, force, and support. Part weight and tolerance are the nodes in the part cluster. We use Rapid Upper Limb Assessment (RULA) [26] to evaluate the ergonomic criteria of each functional action. Force and support criteria are given with a higher importance score if the functional action required force exertion and support on the part. The importance score for the criteria of part weight increases with the weight of the component. Tolerance of the component is classified as rough, fair and tight tolerance. This indicates that high importance score is given to the tight tolerance requirement. We consider the HRC work cell that consists of one human and two robots. Therefore, functional actions can be assigned to either human only, a robot, two robots or a human with a robot.

The relationship between nodes and clusters expresses the influence of each cluster on the other. In the defined network, the criteria in the skill and part clusters affect the assignment of functional action and alternatives. Among the criteria,

tolerance affects force and support. At the same time, part weight affects the ergonomic factor. Each functional action has its assignment to alternatives based on the criteria. The capability of agents to perform the criteria affects the assignment of functional action to the agent. As an example, the human can handle tight tolerance better than the robot, but the human is weak in handling heavyweight. In contrast, the collaboration of two robots can handle heavyweight, but they are less capable of achieving tight tolerance. Thus, we construct the feedback from alternatives cluster to both criteria clusters to convey the different levels in the capability of agents in the functional action assignment. Since there are nine types of functional actions, we repeat the ANP decision model to obtain the ranking of functional action-agent assignment for every functional action. The ranking of agents for each functional action is obtained by normalizing the values of the cells in the limit super matrix. The results for all nine functional actions are summarized in Table 1. We then utilize these results in the final task allocation for the assembly operation and will be discussed in detail in Section 3.3.

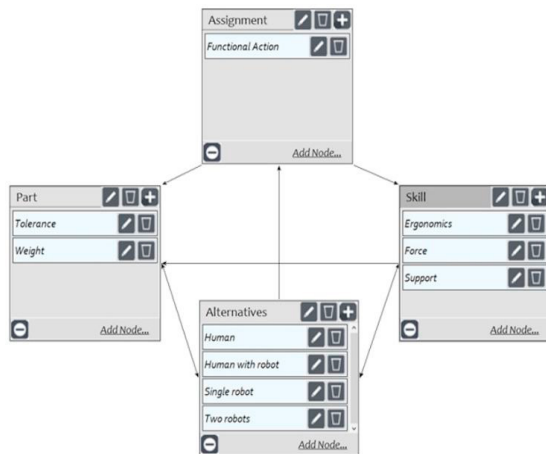


Fig. 3. ANP network for functional action assignment priority.

Table 1. Functional action-agent assignment priority ranking. (Indication: H – Human; R – Robot 1; C – Human with robot 2; J – Both robots)

Code	Functional action	Agent assignment rank			
		1	2	3	4
HR	Lift and position plate with rough tolerance	J	C	H	R
HF	Lift and position plate with fair tolerance	J	C	H	R
HT	Lift and position plate with tight tolerance	C	J	H	R
LF	Pick and locate component with fair tolerance	H	R	C	J
LT	Pick and locate component with tight tolerance	H	R	C	J
LS	Pick, locate and insert screw	H	R	C	J
TS	Tighten screw	R	H	C	J
FS	Insert small component with force exertion	H	C	R	J
FP	Insert plate with force exertion	C	H	J	R

3.3. Functional action allocation for HRC system

We perform the task-agent allocation for the execution of the HRC work cell based on three inputs. These three inputs are functional action precedence from task decomposition (Section 3.1), functional action-agent assignment priority (Section 3.2), and the processing time of each task by each agent during operation. Fig. 4 illustrates the procedures for task-agent allocation. Each functional action in the precedence list is assigned to the priority agent. If the priority agent is unavailable, then the action will be assigned to the second priority agent and so on. We define the last priority agent as the agent that is incapable of executing the action, hence the last priority agent will not be considered. Since we consider an HRC mold assembly cell that consists of one human and two robots, hence there are four alternatives for agents available. In other words, we exclude the fourth priority agent in the assignment. If all the agents are unavailable, we will assign the action to the agent with the earliest completion time. In this paper, we simplify the problem by considering automatic tool change by the robot itself whenever tool change is required. The procedures are repeated until all the functional actions are allocated to agents.

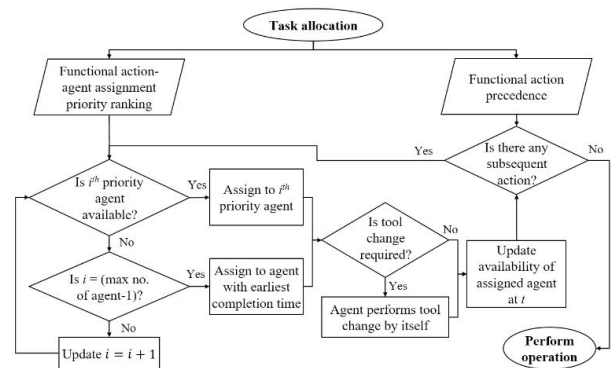


Fig. 4. Task allocation procedure.

4. Implementation of Genetic Algorithm for Task Allocation

GA is implemented in this paper because of the simplicity and efficiency of GA to find an acceptable optimal solution for the multi-objective problem within a reduced computational time. Thus, we implemented GA to achieve the following objectives:

- Minimization of the cycle time per unit mold. Cycle time includes process time, setup time, as well as idle time. The cycle time for each functional action depends on the agent who executes the functional action. Assignment of a functional action to an agent with higher execution capability may require longer processing time due to tool change and speed of the robot. Therefore, we need to achieve optimization between cycle time and the agent's capability.
- Minimization of action assignment to an incapable agent. It is important to eliminate the assignment of functional action to an incapable agent. In other words, it is necessary to

ensure the capability of an agent in executing functional action to avoid robot malfunction and delay in production.

4.1. Chromosome structure

We programmed the GA algorithm for this paper using Python language. Fig. 5 shows the chromosome structure in the GA algorithm. The sequence of the genes in the chromosome represents the sequence of functional actions to be executed. Number of genes equals to the total number of functional actions obtained from the task decomposition. Each gene represents the agent who is allocated to each functional action.

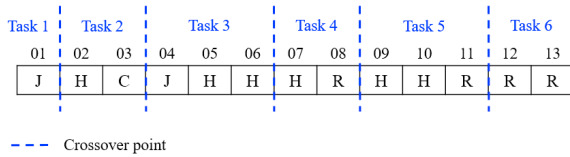


Fig. 5. Chromosome structure.

4.2. Fitness functions

The fitness function f of each chromosome with two objectives described is defined in Equation (1).

$$f = w_t \cdot f_t + w_c \cdot f_c \quad (1)$$

where f_t is the normalized fitness value of minimization of cycle time and f_c is the normalized fitness value of minimization of action assignment to an incapable agent. w_t and w_c are weights of each objective. We define the values of w_t and w_c equals to 0.7 and 0.3 respectively. The lower its value, the higher the level of fitness since all objectives are minimization problems.

The first objective is to minimize the cycle time of assembly operation which is the sum of processing time and tool change time of each functional action in the assembly operation.

$$f_t = \frac{\sum_{i=1}^N (T_i + S_i)}{\sum_{i=1}^N (\max T_i + S_i)} \quad (2)$$

where T_i and S_i are the processing time and tool change time of functional action i respectively and N is the total number of functional actions in the assembly operation. As shown in Equation (2), f_t is defined as the ratio of the sum of T_i and S_i corresponding to the agent assigned to the sum of maximum T_i and S_i for each functional action i .

The second objective is to minimize the assignment of functional action to an incapable agent. We first formulated the Equation (3) to indicate the maximization of the capability of agents. Since the overall objective is a minimization problem, hence, a minus sign is added to convert the maximization of capability agent to our minimization problem.

$$f_c = - \left(\frac{\sum_{i=1}^N C_i}{N} \right) \quad (3)$$

We calculated the capability ratio of each functional action, C_i by using Min-Max feature scaling on the value from ANP analysis. The value C_i must not less than zero ($C_i > 0$) because an agent must have a certain capability level to execute the assigned functional action. The normalized fitness value of the minimization of incapable agent assignment, f_c is the ratio of the sum of capability ratio, C_i for functional actions to the sum of maximum capability ratio which is equal to 1.0, hence it is equal to the total number of functional actions, N .

4.3. Genetic Operators

Tournament selection is used to select fittest candidates from current generation and pass to the next generation. In this paper, we chose five individuals from the population at random. Since it is a minimization problem, hence the best individual from the tournament with the lowest fitness value is selected for crossover. K-points crossover is applied where K is the number of break points between tasks (Fig. 5). Hence, K value as total number of tasks minus one. Then, swap mutation is applied to exchange two genes in the same individual.

5. Case Study and Discussion

5.1. Case study

In this paper, we focus on the HRC application in the two-plates mold assembly (Fig. 6). From the result of task decomposition, the mold assembly consists of two sub-assemblies which are cavity and core sub-assemblies. In this paper, a cavity sub-assembly is used to test the proposed algorithm.

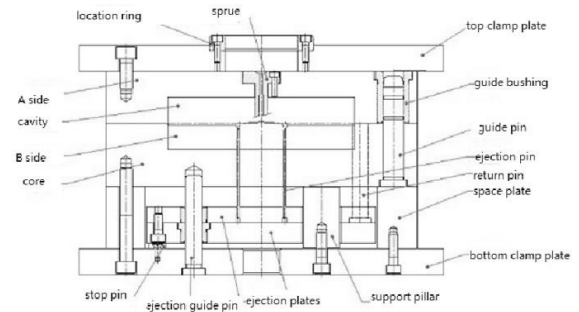


Fig. 6. Structure of two plate mold [27].

From the result of task decomposition, the cavity sub-assembly contains six tasks that decomposed into thirteen functional actions. Thus, the chromosome in the GA algorithm has thirteen genes and the K value of the crossover equals to five. These functional actions are to be executed in a 1H:2R work cell (Fig. 7). The robot used in the simulation is a collaborative robot which has a payload of 5 kilograms and reachability up to 850 millimeters. The yellow circle represents the workspace of the human, and the green and blue circles represent the workspaces of the robot 1 and robot 2 respectively. The overlapped workspaces indicate that the human and both robots can interact and collaborate with each other in the 1H:2R work cell. The human and robot 1 work

individually, and robot 2 collaborates with either the human worker or robot 1. Hence, we can come up with four alternatives of agents that can be assigned to each functional action. The four alternatives are human (H), robot 1 (R), a human with robot 2 (C) and robot 1 with robot 2 (J). In the current manual approach, all the tasks are performed by a human worker in a manual work cell. The human worker arranges the parts in sequence and assembles the parts on the working table. The tools used by human worker are hex key sets for screw tightening and hammer for inserting component. For the robot operation, we can install one of the three types of end effectors on the robot to execute different functional action. The three types of end effectors are vacuum gripper to pick and place flat components such as plate, finger gripper to pick and place small components such as pin and screwdriver.

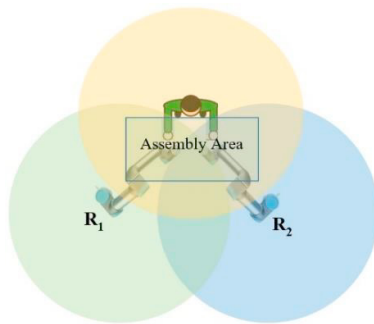


Fig. 7. Proposed one human with two robots (1H:2R) collaboration.

Table 2. Summary of data for process time, tool used and capability ratio for assignment. (Indication: *i* – functional action; FA – type of functional action; H – Human; R – Robot 1; C – Human with robot 2; J – Both robots; V – vacuum gripper; F – finger gripper; S – screw driver)

<i>i</i>	FA	Processing Time (s)				Tool				Capability ratio			
		H	R	J	C	R	J	C	H	R	J	C	
1	HR	10	-	15	12	-	V	V	0.08	0.00	1.00	0.27	
2	LT	10	10	-	10	F	-	F	1.00	0.37	0.00	0.28	
3	FS	20	35	-	30	F	-	F	1.00	0.34	0.00	0.38	
4	HF	15	-	20	17	-	V	V	0.19	0.00	1.00	0.24	
5	LT	10	10	-	10	F	-	F	1.00	0.37	0.00	0.28	
6	FS	20	35	-	30	F	-	F	1.00	0.34	0.00	0.38	
7	LS	20	20	-	30	S	-	S	1.00	0.37	0.00	0.26	
8	TS	40	16	-	30	S	-	S	0.92	1.00	0.00	0.64	
9	LF	8	10	-	10	F	-	F	1.00	0.37	0.00	0.21	
10	LS	10	10	-	20	S	-	S	1.00	0.37	0.00	0.26	
11	TS	20	8	-	20	S	-	S	0.92	1.00	0.00	0.64	
12	LS	20	20	-	30	S	-	S	1.00	0.37	0.00	0.26	
13	TS	40	16	-	30	S	-	S	0.92	1.00	0.00	0.64	

As shown in Table 1, we categorized each functional action into one of the nine functional action types, and each type has its agent assignment priority based on the capability of agents. The fourth-ranked agent is considered as incapable of executing the respective functional action and will be excluded from the alternatives of agents in the proposed algorithm. Hence, there is no input of process time for the fourth-ranked

agent. Since the automatic tool changer is available in the market, hence we assume automatic tool change by the robot itself and define the time for tool change as 20 seconds. The processing time, the tool used, and capability ratio of the functional actions depend on the agents. Table 2 summarizes the data set for those thirteen functional actions in the cavity sub-assembly. The data of processing time for manual operation were collected from a practical mold assembly. Besides, we used Visual Components software to simulate and estimate the processing time for robot and collaborative operations.

5.2. Result and discussion

The result of task allocation for cavity sub-assembly using one human with two robots collaboration work cell is shown in Table 3. The cycle time of the cavity sub-assembly operation in the 1H:2R HRC work cell included the tool change time (20 seconds) is 213 seconds or equal to 3 minutes and 33 seconds. All the selected agents are the first ranked agent with capability weight 1.0. Tool change is performed only once during the assembly operation for task number 8 where the robot changes its end effector from vacuum gripper to screwdriver.

Table 3. Task allocation result. (Indication: *i* – functional action; FA – type of functional action; H – Human; R – Robot 1; C – Human with robot 2; J – Both robots; V – vacuum gripper; F – finger gripper; S – screw driver; TC – tool change)

<i>i</i>	FA	Agent	Processing time (s)	Tool	Capability ratio
1	HR	J	15	V	1.00
2	LT	H	10	-	1.00
3	FS	H	20	-	1.00
4	HF	J	20	V	1.00
5	LT	H	10	-	1.00
6	FS	H	20	-	1.00
7	LS	H	20	-	1.00
8	TS	R	16	S (TC)	1.00
9	LF	H	8	-	1.00
10	LS	H	10	-	1.00
11	TS	R	8	S	1.00
12	LS	H	20	-	1.00
13	TS	R	16	S	1.00

In this paper, we implemented the 1H:2R collaboration instead of a common one human with one robot collaboration because the maximum robot payload of the existing collaboration robot is only up to 15kg. Since the part weight in a mold can exceed 15kg, hence one robot is incapable of handling those heavy parts. We evaluated the proposed one human with two robots (1H:2R) collaboration work cell based on cycle time per unit, ergonomic score and human utilization. Table 5 shows the performance comparison between the 1H:2R HRC work cell with the manual assembly operation. In the 1H:2R work cell, the screw tightening actions were allocated to the robot. Since the robot tightened screws faster than the human, the operation cycle time of 1H:2R work cell was 30 seconds shorter than the manual work cell. In addition to that, we evaluated the ergonomic factor using RULA because only the upper limb motions were performed by the human during the operation. RULA assessment evaluates the score of risk based on the positions of upper limbs and the force or load

required during the task execution. The upper limbs included for assessment are neck, upper and lower arms, wrist and trunk. The base line of the posture is when the human stands straight with arms rest at the sides. The scores of postures are evaluated based on the defined guideline. Higher score is given when the upper limb moves to a greater degree from the base line. RULA assessment defines four level of musculoskeletal disorders (MSD) based on the final score (Table 4). We evaluated every functional action in the video of the mold assembly using the RULA guide. The average RULA score for manual operation is 4.54 which considered as a medium level of MSD risk. The introduction of two robot collaboration in the operation reduces the MSD risk level on the human to a low-level risk level (score = 2.31). In our case study, the functional actions with a very high level of MSD risk (RULA score = 7) in the current manual approach are lifting and positioning plates. These high-risk functional actions are assigned to two robots in the 1H:2R work cell. Furthermore, the repetitive motion such as screw-tightening with a medium MSD risk level (RULA score = 5) is assigned to the robot in the HRC approach. As a result, it eliminates these high MSD risks on the human worker. The human utilization has reduced to 55.4% in the 1H:2R assembly work cell. We calculated the human utilization as the percentage of human working time over the cycle time per unit mold. Higher human utilization indicates longer human working time during the operation. This means that more tasks are assigned to the human worker. In other words, more human attention is required as the number of tasks (workload) increases. Thus, higher human utilization increases the human attention during operation and leads to higher rate of human fatigue or tiredness. The reduction of human utilization in the 1H:2R work cell enables a human worker to maintain his/her performance at steady and optimum conditions for longer working hours since the factor of tiredness or fatigue has been shifted to the robot(s).

Table 4. Level of MSD risk based on RULA score [26].

Score	Level of MSD risk
1-2	Negligible risk, no action required
3-4	Low risk, change may be needed
5-6	Medium risk, further investigation, change soon
6+	Very high risk, implement change now

Table 5. Performance comparison of human utilization and ergonomics factor between manual work cell and proposed one human with two robots (1H:2R) collaboration work cell.

Work cell	Cycle time (minutes/unit)	Average ergonomic score	Human Utilization (%)
Manual	4.05	4.54	100
1H:2R	3.55	2.31	55.4

6. Conclusion and Future Work

This paper proposed a task allocation model for human-robot collaboration mold assembly based on task characteristics and agent capability. We utilized ANP for analyzing characteristics of tasks and agents to define the capability of agents. With the input of the agent's capability and the processing time into the optimization algorithm, this

paper contributed to a human-robot collaboration task allocation model that aims to achieve optimization in the operation cycle time and capability of agent for functional action execution. In addition to that, a human with two robots (1H:2R) configuration was proposed instead of common one human with one robot collaboration cell to expand the variety of part weight and task that can be handled by the HRC cell. From the result of performance comparison, the 1H:2R configuration proposed showed improvement in the cycle time and human utilization when compared to manual work cell. Besides, the reduction of average ergonomic score proved that the injury risk of human can be reduced by implementing the 1H:2R configuration in the cavity assembly.

Since we presented the application of 1H:2R configuration by using a simulated use case, hence we aim to set up an experimental cell to practically test and implement the proposed algorithm in the operation. We will develop a computer vision system in the experimental cell to improve the interaction between the human and robots. Besides, we will define a rule for tool change to minimize non-value added operation. For generalization, we will expand the proposed applications to different types of collaboration modes in one human with two robots collaboration system.

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